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HexiLexi

STREEN AI Companion

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Abstract

Young learners frequently encounter unfamiliar vocabulary that disrupts reading comprehension. Traditional supports, such as static dictionaries, often promote passive learning and result in limited long-term word retention. To address this, we developed HexiLexi, an AI-powered vocabulary support tool integrated into the STREEN digital reading platform. Rather than providing immediate definitions, HexiLexi utilizes inquiry-based prompts driven by a Large Language Model (Gemma 3:12b) to guide children through contextual clues and word morphology.

The application was built as a React component and features four distinct learning modes: Explain, Discover, Synonym, and Word Parts. The tool was evaluated across user-centred sessions with primary school children aged 8 to 10. Quantitative and qualitative results indicate high user engagement, with positive ratings for the system's visual design and interactive functionality. However, the evaluations also revealed a need for adaptive scaffolding, as some AI-generated explanations were perceived as too complex for the target demographic. Ultimately, HexiLexi demonstrates that AI can serve as an effective, guided learning companion to foster word consciousness and active meaning construction in primary education.

Keywords: Artificial Intelligence, Large Language Models (LLMs), Vocabulary Acquisition, Educational Technology, Prompt Engineering, Human-Computer Interaction (HCI), Gamification, Primary Education.

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1. Introduction

1.1 Context

STREEN (Story reading environmental enrichment) is an augmented reading technical framework, which provides a smart environment for story reading. It enriches reading-aloud activities in primary school with images, sounds and light effects, and engaging features which are necessary for fostering long term reading competences, such as vocabulary acquisition, reading fluency, reading comprehension & motivation. To support that approach, digital media is being used. It consists of a tablet app, a projector, a sound system, and smart lights and allows children to collaboratively create digital media that they later use while reading the story aloud in front of an audience. The goal is to increase intrinsic reading motivation of children, by constructing an entertaining reading environment, where children are being motivated to read and enhance their reading skills (Ribeiro et al., 2020). In addition, the results of the studies indicate that advancing the reading instruction with STREEN digital-media technology, integrating sound practices for promoting motivation and text comprehension, can support the development of reading comprehension and motivation in third-grade students.

The pedagogical framework that promotes reading motivation and comprehension while using STREEN is IRIS (Integrated Reading Instruction). It is divided into different supportive instructional elements: first, it focuses on motivation-enhancing instructional practices, designed to support students' psychological needs for relatedness, autonomy, competence, curiosity, and active involvement in reading. Teachers provide engaging and relevant reading materials, offer choice, and encourage collaborative work aimed at mastering reading skills. Transparent goals, constructive formative feedback, and opportunities for self-reflection help students evaluate their successes and failures appropriately. Through scaffolding, teaching and applying reading strategies, IRIS directly strengthens reading self-efficacy and comprehension. Overall, the approach promotes motivation, mastery orientation, and students' recognition of the relevance of reading. Secondly, it's providing a meaningful and authentic context, which consists of the Reader's Theatre, Mini-lessons and STREEN. Our project focuses on SCREEN, so it will be explained in the following what it has to offer: STREEN is designed to support students accomplishing instructional tasks, applying strategies, reading digital texts, reflecting on reading, learning unknown vocabulary, expressing knowledge, documenting their work and performing Reader's Theatre. It can also be used by the teacher to help the children evaluating their progress. The functionality contributes different technological features, which should motivate and help the children with their reading progress:

1. Story Script: Enables visual display of the script with touch-based navigation and supports children by showing line numbers and organizing the text according to roles.
2. Highlight: Children can mark the text to draw attention to the lines they have to read aloud.
3. Modelled Reading: A human voice, which is reading the story. Furthermore, children can decide if they want to hear the whole story or only a part of it. Also, they can tap on a word to hear how the word sounds when read isolated.

4. Word Cloud: Children can select words that are collected and displayed as a word cloud on a classroom projection screen. Selected words are highlighted, and their frequency is visualized through varying font sizes, like a tag cloud.
5. Authoring: This feature allows children to link words from the text to digital media effects by creating interactive illustrations using drag-and-drop tools and predefined multimedia elements, including visuals, light, and sound. The process supports reading comprehension, encourages discussion and collaboration, and provides a product that can be projected and evaluated by the teacher.
6. Triggering: Happens after the authoring process, where all the augmented words appear underlined in the script view. Then they can trigger the chosen words, by tapping the word or sliding their finger below the augmented words.
7. Book cover: Sums up the story in a cover, helping students to understand the text.
8. Record, Listen, Reflect (RLR): This feature enables children to record and review their reading, reflect on their performance, and set improvement goals using guided self-assessment questions.
9. Lexicon: The system includes an embedded lexicon offering definitions for potentially unknown words, while also prompting students to apply contextual strategies, such as rereading the sentence, to determine meaning independently.

1.2 Motivation and Significance of our Project

The Lexicon function was our focus with this project. Our motivation was to not only design a functionality that supports children passively with word definitions, but by actively encouraging them to figure out the word meaning with additional support.

It is common for young learners to encounter unfamiliar vocabulary during reading, which can disrupt comprehension and cause difficulties in the process of meaning-making. Without access to a broad range of words, children are being confronted with different problems such as struggling to keep up with lessons, text comprehension and expressing their ideas clearly (Sahin et al., 2025). This leads to the assumption that schools should place greater emphasis on expanding children's vocabulary. However, Bowers and Kirby state that teachers are not being taught to teach vocabulary in primary schools, so not a lot of time is being dedicated to teaching new words in primary grades. This leads to children developing a vocabulary gap, which will widen over the next years and limit them in their future studies (Neugebauer et al., 2017). Another study proves, that building a strong vocabulary is essential for broader academic success (Foorman et al., 2016), with it being a crucial determinant of reading comprehension (Ricketts, Nation & Bishop, 2007). Beck (2002) & Nation (2013) even state that vocabulary plays a foundational role in the development of the four key language skills: listening, speaking, reading and writing. Moreover, promoting word consciousness can support language and reading outcomes (Scott & Nagy, 2004). To achieve this goal, learning and expanding vocabulary should start early in children's lives, since they must learn many words each school year to be successful readers (Zucker et al., 2021). A strong foundation also ensures that children have the courage to participate in their school lessons. Now with STREEN, we have a Lexicon, which provides immediate definitions of every word.

Nonetheless, some studies show that traditional supports, such as static dictionaries and immediate definitions, promote passive learning and are associated with limited long-term vocabulary retention (Ebbers & Denton, 2008; Loftus-Rattan et al., 2016). Therefore, we had the motivation to develop a platform, that ensures to support children in their reading process without providing them immediate definitions, to foster critical thinking and word consciousness. These are the foundations for lifelong linguistic experience (Ricketts et al., 2007). The project was motivated by the practical opportunities provided by Project TransRegINT, specifically utilizing the LeseLab as a real-world testing environment to observe how children interact with digital reading tools.

It is important to understand how children are expanding their vocabulary in the most effective way. Children learn new vocabulary most effectively with already existing background knowledge or schema (Anderson, 1984; Anderson & Pearson, 1988; Mayer, 2024). According to cognitive theory, new information is kept more successfully when it is meaningfully connected to prior knowledge and presented in relevant contexts (Mayer, 2024). If learners cannot understand, what they are learning, they are unlikely to remember it. Therefore, connecting new concepts to familiar ideas strengthens comprehension and memory (Anderson, 1984). Furthermore, research on effective vocabulary has shown that linking new words to familiar concepts through synonyms and antonyms, as well as using child-friendly definitions, makes the vocabulary terms easier to remember (Beck et al., 2013; Manyak & Kappus, 2021). When introducing new vocabulary, it is beneficial to provide visual representation, use the word in meaningful sentences, and embed it in authentic contexts or stories, as contextualized learning enhances memorability (Loftus-Rattan et al., 2016; Manyak & Kappus, 2021; Marzano, 2009; Yeung et al., 2020; Zucker et al., 2021). Having a clear purpose for reading also improves attention and recall during comprehension (Anderson & Pearson, 1988). Repeated exposure and systematic review are essential for long-term retention, as students need to be taught and retaught vocabulary multiple times (Loftus-Rattan et al., 2016; Manyak & Kappus, 2021; Yeung et al., 2020; Zucker et al., 2021). Interactive review strategies, such as vocabulary games that provide clues through synonyms, antonyms, definitions, or examples, increase exposure and reinforce word knowledge (Manyak & Manyak, 2021). Studies have also shown that active engagement is more effective than passive exposure. Providing multiple opportunities for discussion and practice with new words promotes deeper understanding than offering definitions alone (Ebbers & Denton, 2008). Also, engaging children in collaborative activities that incorporate taught vocabulary improves general vocabulary development (Wasik & Hindman, 2020). Moreover, it is being promoted to track and celebrate children's use of newly learned words to further strengthen vocabulary acquisition (Zucker et al., 2019). A comprehensive vocabulary program should include wide reading, direct instruction of selected words, explicit teaching of word-learning strategies, and fostering word consciousness (Graves, Schneider, & Ringstaff, 2018; Blachowicz et al., 2013). Word consciousness, which is defined as an interest in and motivation to learn and use new words, supports sustained vocabulary growth (Graves & Watts-Taffe, 2008). Encouraging children to actively attend to and search for new words, for example by prompting them to listen for unfamiliar vocabulary during read-aloud, further enhances awareness and engagement (Zucker et al., 2021).

Therefore, we developed HexiLexi, an AI-based vocabulary support function integrated into the *STREEN* reading platform, designed to foster vocabulary learning and support active meaning construction. Rather than providing immediate definitions, HexiLexi is designed to use AI-driven, inquiry-based prompts that encourage children to reflect on word morphology and contextual clues. Through this guided process, initial uncertainty becomes a chance for exploration, supporting children in acquiring strategies for independent vocabulary learning.

1.3 Goals

The primary goal of HexiLexi is to support vocabulary development while fostering the critical thinking and word consciousness necessary for independent reading. By expanding the children's vocabulary, a positive impact of word consciousness on language and reading (Scott & Nagy, 2004). Also, by using the creative term of the word wizard "HexiLexi", attempts are being made to help building excitement for finding new words (Bowers & Kirby, 2010).

Based on the cognitive theory (Anderson, 1984; Mayer, 2024), HexiLexi aims to promote deep processing by encouraging children to connect new vocabulary to prior knowledge rather than receiving passive definitions. We want to support meaningful learning through schema activation and contextual reasoning. Drawing on research on word-learning strategies (Graves et al., 2018; Manyak & Kappus, 2021), HexiLexi aims to teach children how to infer word meanings using: contextual clues, morphological awareness, synonyms and antonyms. The long-term goal is to foster strategic, self-regulated vocabulary learning. Inspired by Graves & Watts-Taffe (2008) and Scott & Nagy (2004), HexiLexi seeks to cultivate curiosity and motivation toward words. By presenting vocabulary exploration as engaging (e.g., through the "word wizard" identity), the tool aims to promote interest in discovering and using new words. Research shows that passive definition delivery leads to limited retention (Ebberts & Denton, 2008; Loftus-Rattan et al., 2016). Therefore, we aim to replace definitions with guided inquiry, encourage reflection before revealing meaning and increase cognitive effort to improve long-term retention. Because studies have shown that vocabulary must be encountered multiple times (Loftus – Rattan et al., 2016; Zucker et al., 2021), our tool revisits unfamiliar words across contexts, encourages repeated engagement and reinforces learning through interaction. Since vocabulary knowledge is a key determinant of reading comprehension (Ricketts et al., 2007; Foorman et al., 2016), HexiLexi aims to reduce comprehension breakdowns caused by unfamiliar words while still maintaining cognitive engagement. To carry on IRIS framework, we want HexiLexi to be a tool, that children use to explore more independently. Also, it should support competence, by guided success experience and curiosity, with the help of inquiry-based prompts. Ultimately, HexiLexi aims to equip children with transferable vocabulary-learning strategies that extend beyond the *STREEN* environment, fostering long-term academic and linguistic development.

2. Methodology

2.1 Research Context

The present work was carried out as part of the development project HexiLexi, a digital learning tool designed to support vocabulary development and semantic elaboration in primary school children. Rather than being conducted as a stand-alone experimental study, the data collection was embedded within an ongoing design and refinement process. The primary purpose was to gather structured feedback and usage-related insights to inform further development of the tool. HexiLexi was tested across four iterative development phases in two primary schools. During these phases, children interacted with successive versions of the tool under classroom conditions. The project aimed to better understand how children engage with the system, how they process and explain vocabulary, how motivated they feel when working with the tool, and how they evaluate its design and functionality.

To support these goals, a combination of standardized motivational assessment, structured vocabulary activities, and participatory feedback methods was employed. All activities took place during regular school hours in authentic classroom settings. This approach ensured that feedback and observations reflected naturalistic usage conditions rather than artificial testing environments.

2.2 Participants

Participants were primary school children in the age from 8-10 recruited from two cooperating elementary schools. Participation was voluntary. Institutional approval and parental consent were obtained prior to data collection. All data were anonymized before analysis. The user-centered school sessions were facilitated through the LeseLab initiative, which provided the structured classroom conditions necessary for our research.

2.3 Instruments and Measures

2.3.1 Situational Motivation

Children's task-specific motivational state was assessed using the Fragebogen zur Erfassung aktueller Motivation (FAM; Rheinberg, Vollmeyer, & Burns, 2001). The FAM is a state-based self-report instrument designed to measure current motivation in learning and achievement contexts. The questionnaire consists of 18 items rated on a 7-point Likert scale (1 = does not apply at all; 7 = fully applies). It measures four motivational dimensions:

- *Interest*
- *Challenge*
- *Probability of Success*
- *Fear of Failure*

Example items include:

“I like tasks such as these riddles and puzzles” (Interest),
 “The task is a real challenge for me” (Challenge),

“I believe I am capable of handling the difficulty of this task” (Probability of Success), and “I am somewhat afraid that I might embarrass myself” (Fear of Failure).

Scale scores were calculated by averaging items belonging to each subscale following the procedure described by Rheinberg et al. (2001).

2.3.2 Vocabulary Task and Lexical Processing Assessment

Vocabulary work was based on short narrative texts provided in printed form. Children read the story and independently marked words they did not understand. These self-identified unfamiliar words served as the basis for subsequent semantic elaboration using HexiLexi. HexiLexi was used to support structured exploration and construction of word meanings. Children documented elaborated meanings on worksheets, allowing externalization of their semantic representations.

To further examine lexical knowledge and explanation strategies, a structured word explanation activity inspired by the game Taboo was implemented. In this task, children explained selected target words while being restricted from using closely related or obvious cue words. This method elicited semantic feature activation, definitional strategies, functional descriptions, and categorical reasoning. Observations focused on how children structured explanations, selected defining attributes, and adapted their communicative strategies under lexical constraints.

2.4 Participatory Design Feedback (Post-It Method)

User-centered feedback on HexiLexi was collected using a structured ‘Post-It’ method in small groups. Children generated written suggestions and reflections, which were subsequently clustered thematically. The feedback addressed three main domains:

- *Gamification elements (e.g., collectables, reward systems, avatars, game mechanics)*
- *Guided interaction and conversational structure*
- *Response modes and interaction formats*

This participatory approach provided qualitative insights into usability, engagement factors, and desired functional adaptations. The method allowed systematic collection of design-related suggestions directly from end users.

2.5 Evaluation of Design and Functionality

Children’s evaluation of HexiLexi was assessed using a ‘Smileyometer’, a visual rating scale developed for child–computer interaction research (Read & MacFarlane, 2006). The instrument consists of a sequence of facial expressions ranging from very unhappy to very happy and enables age-appropriate evaluative judgments. In this study, the Smileyometer was used to assess:

- *children’s evaluation of the visual and structural design of the tool, and*
- *their perception of how well the functions operated.*

Responses were recorded numerically based on the selected facial expression and analyzed descriptively. The visual format minimized linguistic demands and supported reliable affective evaluation by primary school children.

2.6 Data Analysis

Quantitative data from the FAM were analyzed by computing mean scores for each motivational dimension. Smileyometer responses were converted into numerical values for descriptive statistical analysis.

Qualitative data from the word explanation task and the Post-It feedback were analyzed using thematic categorization. Lexical explanation strategies were coded according to semantic and communicative features. Design feedback was clustered into usability and gamification-related categories to inform subsequent development phases of HexiLexi.

3. Implementation

The implementation phase of HexiLexi translates the conceptual goals of inquiry-based vocabulary learning into a functional, interactive digital application.. This section details the system architecture, the integration of the Large Language Model (LLM), the highly iterative prompt engineering and design process, the logic behind the core learning modes, and the solutions developed to overcome technical challenges during development and user testing.

3.1 System Architecture and Integration

The initial technical approach for HexiLexi involved developing a standalone full-stack application, complete with a dedicated backend (server.js) to handle API requests and logic. This approach was chosen to build foundational knowledge, as the development team was relatively small (six members) and navigating LLM integration for the first time. The first goal was to set up a foundation where we can interact with the LLM API reliably and develop a basic UI for the application.

However, as the project progressed, the architecture was heavily refactored. To fulfil the core requirement of serving as a companion tool within the existing STREEN digital reading platform, the traditional backend was stripped away. HexiLexi was reconstructed entirely as a modular React component. This shift allowed the application to run entirely in the browser, communicating directly with the LLM API via a dedicated frontend service (llmService.js). By converting the system into an easily importable React component, it could be seamlessly embedded into STREEN's digital reading environments without requiring complex backend integrations or altering STREEN's existing server infrastructure.

3.2 Technology Stack

The frontend was built using React (version 18.2) alongside the Create React App toolchain. For the user interface, Ant Design (version 5.9) was utilized to rapidly prototype and implement clean, accessible UI components. This was particularly important given the absence of dedicated design students on the team.

Network requests to the AI model were handled using Axios, while native browser capabilities were leveraged for accessibility features. Specifically, the Web Speech API was integrated to allow hands-free voice input. The speech recognition was strictly configured to German (de-DE) to match the target demographic, processing spoken input and passing it through a custom sanitization and formatting function (tidySentence) to ensure the text was properly capitalized and punctuated before being evaluated or sent to the LLM.

User progress, including accumulated points, current level, login streaks, and the personalized dictionary of learned words, was implemented using browser cookies. This allowed for persistent gamification states across reading sessions without requiring a database. To ensure application security, all user inputs are strictly processed through a custom 'sanitizeInput' function. This mechanism strips potentially malicious HTML tags and script content while preserving necessary German characters, mitigating the risk of Cross-Site Scripting (XSS) attacks before the data is evaluated or transmitted to the LLM.

3.3 Language Model Integration and Configuration

The core intelligence of HexiLexi is powered by a Large Language Model provided by the university's AIS Lab. Initially, the system was built around the Llama 3.3 model. However, early testing revealed significant latency issues, with the model averaging approximately 30 to 45 seconds to generate a response for a single query. This delay was unacceptable for an application targeting young children, as it severely disrupted engagement, and proved to be difficult for quick testing changes to the prompts.

To resolve this, the system was migrated to the Gemma 3:12b model hosted via the AIS Lab's Ollama API server. The transition to Gemma 3:12b provided a drastic improvement in response times while maintaining high proficiency in the German language and instructional dialogue. Looking toward the future, the AIS Lab is currently testing the larger Gemma 3:27b model, and the HexiLexi architecture has been designed to easily scale and adapt to this newer model simply by updating the configuration variables in config.js once it becomes available. We have yet to analyse the impact of running bigger, faster, or more intelligent models in HexiLexi since it did not fit our time constraints nor did we have the resources to run them locally, however our research and intuition indicates that they will further improve the quality of responses and interactivity between HexiLexi and children.

3.4 Prompt Engineering and Iterative Development

The prompt engineering process was one of the most resource-intensive phase of the project, spanning approximately three months of continuous iteration. Because this was the team's first experience working with LLMs, the process required extensive foundational research. The team studied pedagogical literature on how to communicate with children in a teaching context, the syntactical elements of words in sentences, and general prompt engineering best practices. We took this learning step into a more practical direction by conversing directly with Cloud LLM models as well as the local Ollama model, to figure out the constraints and respective limits of the models we were working with, and how to structure our prompts for better instructions and guidance.

HexiLexi was originally just a generic wizard, but was then given a distinct persona, a friendly, magical character, to match the avatar and to make the learning experience approachable. We received positive reception on the rebranding of the application with the new mascot and name from children during our second LeseLab testing session. Originally, we worked on simplifying our prompts and to send one large payload to the AI on every request. However, after further testing and research, the prompts were engineered using a two-step "Architect and Builder" pipeline implemented in `llmService.js`:

The Architect: When a child clicks an unfamiliar word, the Architect prompt analyzes the word within the context of the specific sentence. It generates a structured JSON output containing the exact meaning, contextual hints, false options for multiple-choice questions, and synonyms.

The Builder: The Builder prompts use the data generated by the Architect to manage the actual conversation flow. Depending on the active learning mode, the Builder constructs the specific questions and dialogue lines that the child sees. Splitting the original, large and bulky prompt into two well-structured "personalities", we were able to finally achieve consistent and expected results from the AI without it losing the instructions or context of the current scenario.

Following the initial prompt design, the team conducted continuous testing sessions using various stories and target words, both locally and on STREEN. The results were documented, analyzed for logical failures or inappropriate AI tone, and reviewed with the project supervisor. This highly iterative loop resulted in a highly satisfactory state for the "Erklären" (Explain) mode. While the other discovery-based modes are functional and effective, the team notes that an additional month of dedicated prompt tuning could further optimize their conversational fluidity and achieve more accurate results.

3.5 Learning Modes and Flow Logic

We implemented gamification into the application for the goal of improving the engagement rate of children with the application. This included rewarding various amounts of points for

each mode, depending on the difficulty imposed by such mode, and allowing the children to unlock different ranks as they “level up” by accumulating points. This was implemented as a clean, minimal and interactive “XP” bar in the UI.

HexiLexi features four distinct learning modes, each programmed with specific conversational state machines in the frontend:

Erklären (Explain): A direct explanation mode that utilizes the Architect's analysis to provide an immediate, age-appropriate definition, with an option for the child to ask for further elaboration. This mode proved to be the most reliable and perform the best.

Entdecken (Discover): This mode operates on a strict multi-phase state machine (maximum 8 turns). Phase 1 asks contextual questions (Who? Where? What is happening?). Phase 2 asks narrowing questions to connect the context to the unknown word. Phase 3 concludes with a multiple-choice confirmation of the meaning. This mode went several iterations over the development periods, including a different name and structure (Riddle mode). We were finally able to achieve an acceptable level of response, but this mode has the potential to be further improved through more testing and iterative development.

Synonym: The LLM guides the child to guess the word's meaning by providing similar words. In future, HexiLexi can be linked with a local or online dictionary/thesaurus API to provide concrete and reliable synonyms to make this mode extremely competent.

Wort-Teile (Word Parts): Designed for compound words (Komposita). The system relies on a heuristic detection function (`detectWordTypes`). Because the LLM occasionally struggled to correctly identify composite words, the team implemented a heuristic override list (`KNOWN_COMPOSITES`) and regex patterns in the frontend to forcefully identify common compound words (e.g., "Sonnenblume", "Zaubertrank"). The conversation logic then guides the child to define the two individual parts before combining them to understand the full word. The composite word detection can be faulty at times, with our testing providing an average of 90% accuracy in detecting composite words and may require a more intelligent model or human intervention to achieve complete accuracy.

To translate the LLM's raw text into a polished user interface, HexiLexi employs custom parsing functions. In the frontend service, a `parseResponse` function utilizes regular expressions to cleanly separate the conversational dialogue from the dynamically generated multiple-choice options appended by the LLM. On the rendering side, a `renderWithFormatting` algorithm processes the dialogue to dynamically render markdown-style bolding and isolate trailing emojis, ensuring the chat bubbles remain visually engaging and readable for children.

Additionally, a custom ‘SkeletonLoader’ component visually mimics a typing indicator to provide immediate feedback while the AI processes responses.

3.6 Frontend-Based Answer Evaluation

During early development, the application relied entirely on the LLM to grade the children's typed or spoken answers. However, testing revealed that the LLM was unreliable as a consistent grader; it would sometimes accept completely wrong answers or reject correct but poorly phrased answers or would forward system prompts and inject system instructions into the responses conveyed to the children.

To ensure pedagogical accuracy, the evaluation logic was moved entirely to the frontend. A custom ‘evaluateAnswer’ algorithm was developed in llmService.js. When a child submits an answer, the algorithm compares the user's input against the correct definition and synonyms provided by the Architect's initial JSON analysis.

The algorithm checks for:

- *exact string matches*
- *"contains" matches (if the child's sentence contains the core correct word)*
- *fuzzy matching using the Levenshtein distance algorithm. By calculating a string similarity score, the system allows for minor spelling mistakes (a common occurrence with 8-to-10-year-olds) if the similarity meets the configured threshold.*

Only after the frontend algorithm accurately evaluates the answer is the result passed to the Builder prompt to generate the corresponding praise, correction or final dialogue.

3.7 Overcoming Technical Challenges: Rate Limiting and UI Locking

The transition from controlled testing to actual classroom deployment highlighted a critical flaw in the initial application state management. During the final user-centered school session (Session 3), the application crashed. The young participants, eager to interact with the gamified elements, repeatedly spammed the mode buttons and submitted multiple requests simultaneously. This overwhelmed the LLM API and caused the application to lock up, ending the testing session prematurely.

To resolve this oversight, we implemented a robust ‘RequestManager’ class in the LLM service. This function handles request queueing and strict rate-limiting.

Alongside the backend queue, strict UI locking mechanisms were introduced in App.js. When a word is clicked or an answer is submitted, the frontend immediately sets a boolean loading state ‘(setLoading(true))’ and registers a ‘thinkingStartTime’. This state disables all interactive elements, including word clicking, mode switching, and text input, until the active API request fully resolves. If the API takes longer than the configured timeout, an ‘AbortController’ terminates the request, displays a friendly timeout error ("Das dauert zu lange..."), and safely

unlocks the interface. This combination of frontend UI locking and backend queue management ensured system stability in all subsequent tests.

4. Results

4.1 Overview of Data Collection and Data Completeness

HexiLexi was evaluated across four school-based sessions with children aged 8–10 years. Quantitative data were collected using structured questionnaires covering Function (*F-items: F1–F7*) and Design (*G-items: G1–G7*). In Session 4, additional Motivation items (*M1–M3*) were included. Qualitative feedback was collected in written form during Sessions 2 and 4.

All questionnaire items were rated on a 7-point semantic differential scale where 1 represents the left anchor and 7 represents the right anchor. For example, F1 ranged from "Easy" (1) to "Difficult" (7), F4 ranged from "Exciting" (1) to "Boring" (7), G3 ranged from "Entertaining" (1) to "Not entertaining" (7), and G7 ranged from "Neat" (1) to "Chaotic" (7).

Due to technical problems with HexiLexi, no quantitative questionnaire results are available for Session 3; therefore, Session 3 is excluded from all numerical analyses. Descriptive statistics are reported for Session 1 ($N = 13$), Session 2 ($N = 13$), and Session 4 ($N = 12$). For all three sessions, no missing values were recorded per item ($Fehlend = 0$).

4.2 Quantitative Results by Session

4.2.1 Session 1 ($N = 13$)

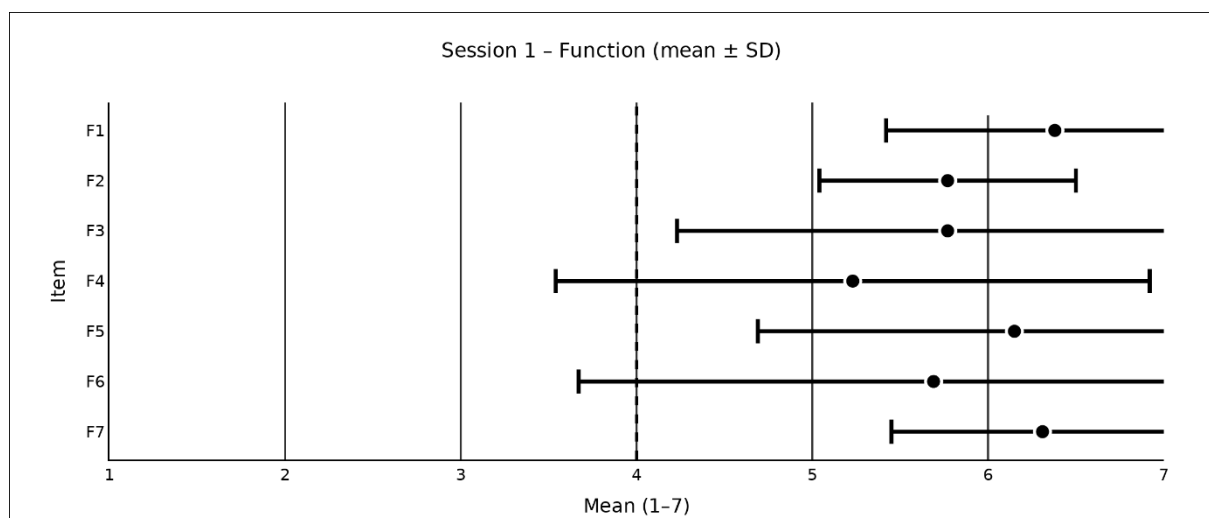


Figure 1: Session 1 Function Items (F1–F7)

F1 (Easy–Difficult), F2 (Good for learning–Bad for learning), F3 (Motivates me–Doesn't motivate me), F4 (Exciting–Boring), F5 (Functions well–Functions badly), F6 (Functions fast–Functions slowly), F7 (Understandable–Unclear).

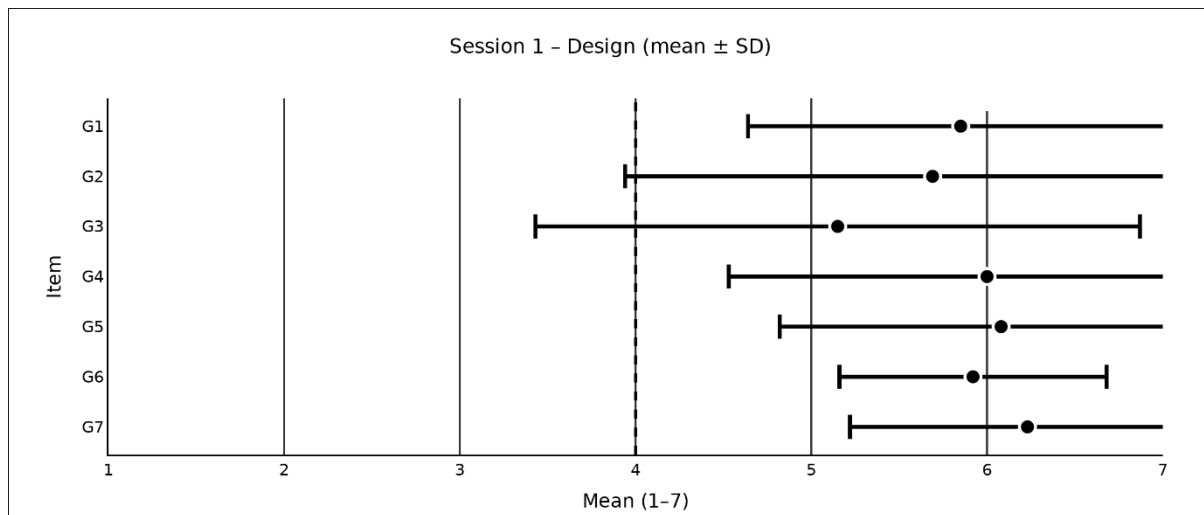


Figure 2: Session 1 Design Items (G1-G7)

G1 (Fun-Boring), G2 (Well-constructed - Badly-constructed), G3 (Entertaining - Not entertaining), G4 (Beautiful - Not beautiful), G5 (Happy-Sad), G6 (Colourful - Simple), G7 (Neat - Chaotic).

In Session 1, function ratings showed high means across most items. F1 (easy-difficult) and F7 (understandable-unclear) had the highest means ($M = 6.38$ and $M = 6.31$, respectively), with relatively low standard deviations ($SD = 0.96$ and $SD = 0.86$), indicating strong clustering of responses toward the "difficult" and "unclear" anchors. F5 (functions well-functions badly) also showed a high mean ($M = 6.15$, $SD = 1.46$). F4 (exciting-boring) had a lower mean ($M = 5.23$, $SD = 1.69$), indicating a tendency toward the "boring" anchor with notable variability. F6 (functions fast-functions slowly) showed the largest variability among function items ($SD = 2.02$).

Design ratings were generally positive with respect to the left anchors (e.g., fun, beautiful, cheerful, neat). G7 (neat-chaotic) showed the highest mean ($M = 6.23$, $SD = 1.01$), suggesting responses clustered toward the "chaotic" anchor, though the low SD indicates consistency. G5 (happy-sad) and G4 (beautiful-not beautiful) also showed high means ($M = 6.08$ and $M = 6.00$, respectively). G3 (entertaining-not entertaining) had the lowest design mean ($M = 5.15$, $SD = 1.72$), suggesting responses shifted toward "not entertaining." G6 (colourful-simple) had the lowest standard deviation among design items ($SD = 0.76$), indicating tight clustering of responses.

4.2.2 Session 2 (N = 13)

In Session 2, F5 (functions well-functions badly) continued to show the highest mean ($M = 6.23$, $SD = 0.83$), with the lowest standard deviation among all function items indicating strong consensus that the system's technical functioning was perceived consistently. F2 (good for learning-bad for learning) also showed a relatively high mean ($M = 5.77$, $SD = 1.59$). In contrast F4 (exciting-boring) had the lowest mean ($M = 4.15$, $SD = 1.91$), suggesting a shift toward the "boring" anchor with increased variability. F3 (motivates me-doesn't motivate me) showed a mean close to the scale midpoint ($M = 4.54$, $SD = 2.22$), indicating greater heterogeneity in motivation perceptions.

Design items in Session 2 displayed variable patterns. G7 (neat–chaotic) showed the highest mean ($M = 6.38$, $SD = 0.87$), consistent with Session 1. G5 (happy–sad) and G4 (beautiful–not beautiful) also showed high means ($M = 6.00$ and $M = 5.92$, respectively). G3 (entertaining–not entertaining) had the lowest design mean ($M = 4.31$, $SD = 1.97$), indicating a stronger shift toward "not entertaining" compared to Session 1. G1 (fun–boring) and G2 (well constructed–badly constructed) showed higher standard deviations ($SD = 2.06$ and $SD = 1.83$), suggesting more diverse responses on these dimensions.

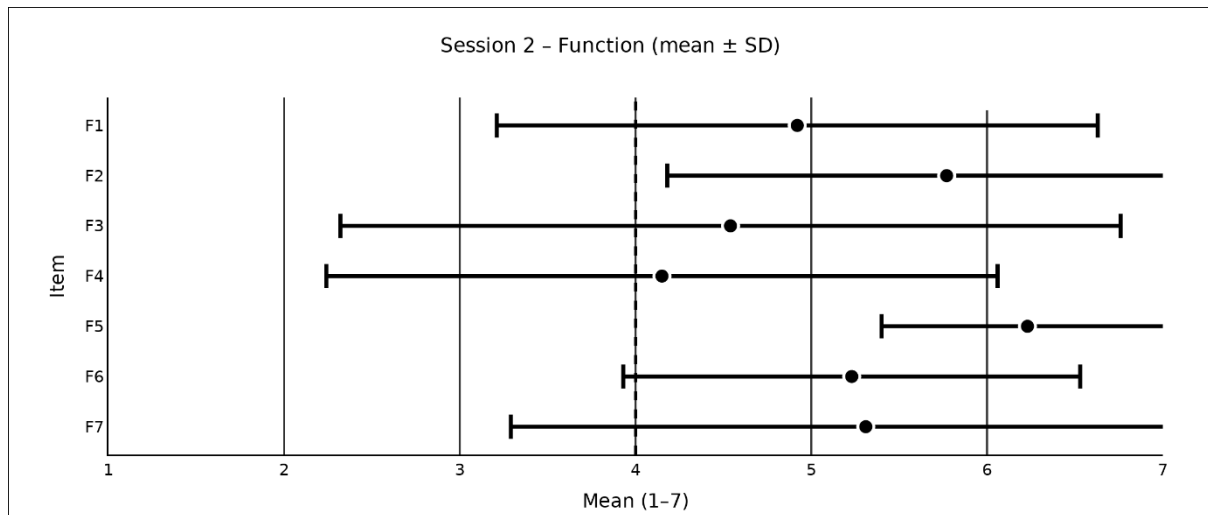


Figure 3: Session 2 Function Items (F1–F7).

F1 (Easy–Difficult), F2 (Good for learning–Bad for learning), F3 (Motivates me–Doesn't motivate me), F4 (Exciting–Boring), F5 (Functions well–Functions badly), F6 (Functions fast–Functions slowly), F7 (Understandable–Unclear)

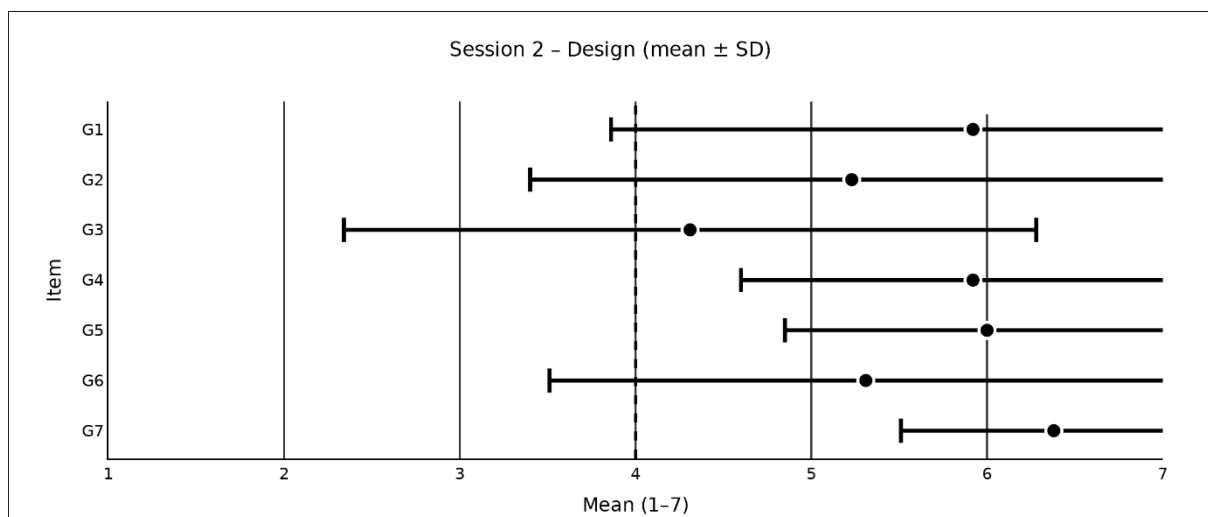


Figure 4: Session 2 Design Items (G1–G7)

G1 (Fun–Boring), G2 (Well constructed–Badly constructed), G3 (Entertaining–Not entertaining), G4 (Beautiful–Not beautiful), G5 (Happy–Sad), G6 (Colorful–Simple), G7 (Neat–Chaotic).

4.2.3 Session 3

No quantitative results are available for Session 3 due to technical issues of HexiLexi; therefore, no statistical analysis was conducted for this session.

4.2.4 Session 4 (N = 12)

In Session 4, F5 (functions well–functions badly) maintained a high mean ($M = 5.83$, $SD = 1.75$) though slightly lower than Sessions 1 and 2. F1 and F7 both showed means around 5.58, with higher standard deviations ($SD = 2.15$ and $SD = 1.73$, respectively) compared to earlier sessions, indicating increased variability. F3 (motivates me–doesn't motivate me) had the lowest function mean ($M = 4.08$, $SD = 2.50$) with the highest standard deviation among all function items, reflecting substantial disagreement in motivation perceptions. F6 (functions fast–functions slowly) also showed high variability ($SD = 2.66$) and a lower mean ($M = 4.17$).

Design items in Session 4 displayed more diverse response patterns than in Sessions 1 and 2. Means ranged from 4.50 to 5.75, with G2 (well-constructed–badly constructed) showing the highest mean ($M = 5.75$, $SD = 1.66$). Multiple design items exhibited high standard deviations (G6 $SD = 2.59$; G3 $SD = 2.41$; G7 $SD = 2.11$; G1 $SD = 2.34$), and minima of 1 appeared for most items except G2 ($Min = 2$), indicating broader disagreement among participants.

Motivation items (M1–M3) showed means near the midpoint ($M1 = 4.83$, $M2 = 4.92$, $M3 = 4.25$) with high variability ($M2\ SD = 2.64$; $M3\ SD = 2.18$; $M1\ SD = 1.64$). These patterns indicate heterogeneous motivation experiences across participants.

4.3 Cross-Session Quantitative Patterns

Across Sessions 1, 2, and 4, F5 (functions well–functions badly) consistently showed relatively high means: Session 1 $M = 6.15$ ($SD = 1.46$), Session 2 $M = 6.23$ ($SD = 0.83$), and Session 4 $M = 5.83$ ($SD = 1.75$). Session 2 exhibited the tightest clustering for this item ($SD = 0.83$), suggesting strong consensus on technical functioning.

Items related to engagement and motivation (F3, F4) showed declining means across sessions. F4 (exciting–boring) decreased from Session 1 ($M = 5.23$) to Session 2 ($M = 4.15$) to Session 4 ($M = 4.67$), with increasing standard deviations ($SD = 1.69, 1.91, 2.35$, respectively). F3 (motivates me–doesn't motivate me) followed a similar trend: Session 1 $M = 5.77$ ($SD = 1.54$), Session 2 $M = 4.54$ ($SD = 2.22$), Session 4 $M = 4.08$ ($SD = 2.50$). The increasing dispersion in later sessions indicates growing heterogeneity in how children experienced motivation and excitement.

Design item G7 (neat–chaotic) showed a consistent pattern of high means in Sessions 1 and 2 ($M = 6.23$, $SD = 1.01$; $M = 6.38$, $SD = 0.87$) but declined in Session 4 ($M = 5.08$, $SD = 2.11$) with substantially increased variability. G3 (entertaining–not entertaining) displayed lower means across all sessions (Session 1 $M = 5.15$; Session 2 $M = 4.31$; Session 4 $M = 4.83$) with consistently high standard deviations (ranging from 1.72 to 2.41), indicating persistent variability and disagreement on entertainment value.

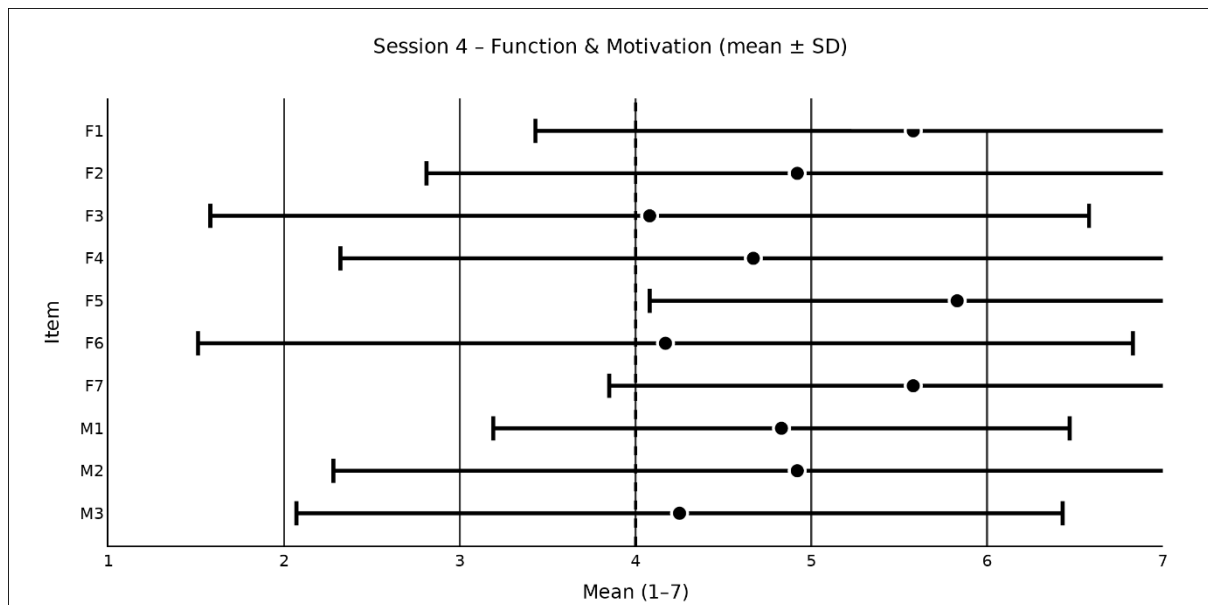


Figure 5: Session 4 Function Items (F1–F7) & Motivation (M1–M3)

F1 (Easy–Difficult), F2 (Good for learning–Bad for learning), F3 (Motivates me–Doesn't motivate me), F4 (Exciting–Boring), F5 (Functions well–Functions badly), F6 (Functions fast–Functions slowly), F7 (Understandable–Unclear), M1 (Determined to make effort), M2 (Motivated today), M3 (Don't feel like making effort).

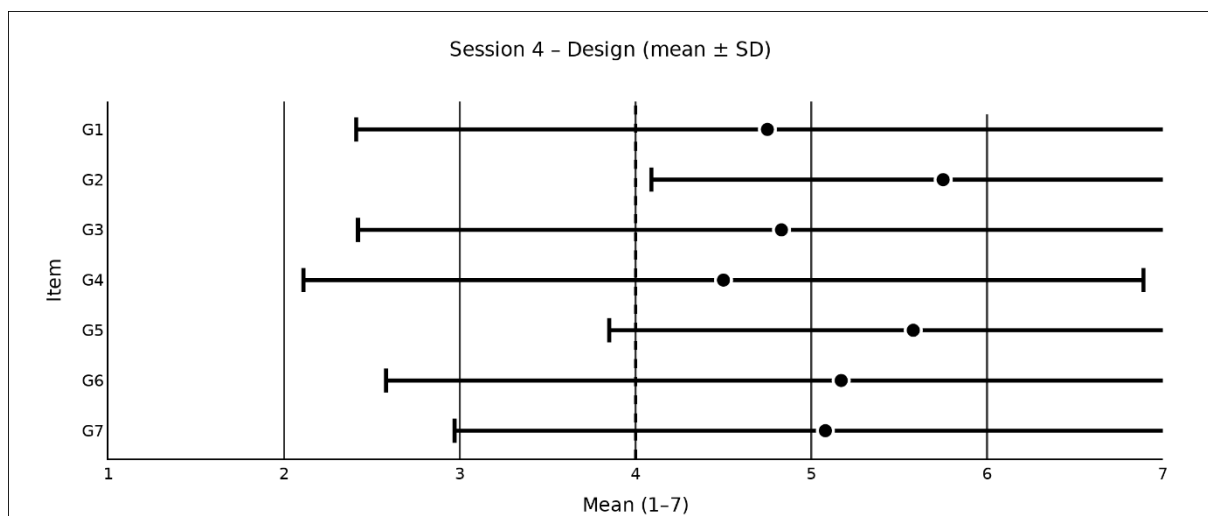


Figure 6: Session 4 Design Items (G1–G7)

G1 (Fun–Boring), G2 (Well-constructed–Badly constructed), G3 (Entertaining–Not entertaining), G4 (Beautiful–Not beautiful), G5 (Happy–Sad), G6 (Colorful–Simple), G7 (Neat–Chaotic).

4.4 Qualitative Results

4.4.1 Positive Feedback

Children reported liking the design and font. Several comments indicated that HexiLexi was motivating and fun. Children appreciated that words were explained and stated that everything was understandable and enjoyable. The points system was also mentioned as a positive feature. These positive comments align with moderate-to-high means observed for design items (G1–G7) in Sessions 1 and 2.

4.4.2 Identified Problems in Explanations

A major qualitative theme concerned explanation quality and clarity. Multiple children reported that explanations were sometimes too long and required too much reading. Several comments indicated that the language level was too difficult and that explanations were hard to understand "if you don't know the words." Participants also noted grammar issues, specifically that explanations did not always follow correct German grammar rules and did not make sense grammatically.

Children criticized the inclusion of irrelevant details. Feedback explicitly mentioned that "fun facts" were unnecessary and caused additional confusion. Participants suggested that explanations should be short, simple, and fact-based. For animal-related words (e.g., "Eidechse"), children recommended that explanations include observable features (such as colour and body structure), habitat, and common knowledge facts to help identify the concept.

Feedback also highlighted mode-specific instruction problems. In riddle mode, the system sometimes offered word options that did not match the target word well. For example, suggested verbs (e.g., "Springen, Fliegen, Kriechen" for a lizard) were noted as not fitting the context, and participants suggested more appropriate alternatives (e.g., "kriechen, rennen, klettern"). In synonym mode, children suggested that the system should present synonyms more directly rather than asking riddle-like questions. For compound mode, feedback indicated that the system should explicitly guide children to split compound words into meaningful parts rather than using a riddle-like format.

4.4.3 Contextual and Narrative Issues

Qualitative feedback revealed inconsistencies in story integration. Children reported that the system sometimes changed character names or assigned gender randomly (e.g., introducing names like "Maya" or "Lina" for the main character). These inconsistencies were experienced as confusing. Participants requested that the system use consistent story character names and maintain appropriate gender consistency.

Explanations sometimes did not align with the intended meaning of a word within the story context. One example involved the word "Erde," where feedback indicated that the explanation did not adequately address the contextual meaning within the story sentence. These observations suggest that children expected explanations to be semantically and narratively coherent with the story situation.

4.4.4 Interaction and Technical Issues

During one session, a technical vulnerability was observed: repeated rapid button presses triggered multiple simultaneous API requests resulting in system crashes. This finding led to implementation changes restricting multiple request triggers. Additional usability issues included words remaining clickable after explanations were completed, which children found confusing, and cases where answer validation was incorrect (e.g., plausible answers marked as wrong). These issues directly impact user experience and interaction flow.

5. Conclusion and Future Work

5.1 Conclusion

The project successfully demonstrated that AI-supported vocabulary assistance can be integrated into a reading environment in a meaningful and child-friendly way. Across three user-centred school sessions with third and fourth-grade students, HexiLexi was iteratively tested, evaluated, and improved. Quantitative findings showed high ratings in system functionality ($M = 6.23$), perceived visual appeal (e.g., cheerful: $M = 6.00$; neat: $M = 6.38$), and general usability. These results indicate that children perceived the tool as technically reliable, visually engaging, and largely intuitive to use.

At the same time, the relatively high standard deviations in areas such as comprehensibility and motivation reveal important insights. While many learners benefited from the explanations and learning modes, others found certain vocabulary items or AI-generated responses difficult. This variation highlights both the heterogeneity of learners in primary school contexts and the need for further personalization within AI-driven educational tools.

Qualitative feedback further reinforced these findings. Children described HexiLexi as fun, motivating, and visually attractive. The different learning modes; Explain Mode, Discovery Mode, Synonym Mode, and Word Parts Mode; were perceived as helpful in understanding unfamiliar words. Particularly in later sessions, where explanatory strategies were reapplied and group-based challenges were introduced, students showed increased engagement and collaborative problem-solving behaviour.

However, the project also revealed several limitations. Technical instabilities during Session 3 prevented a full pre–post comparison of vocabulary development. Moreover, motivation was not assessed using a validated measurement instrument, which limits the strength of conclusions regarding long-term motivational impact. In addition, some AI-generated explanations required prompt refinement to ensure clarity, age-appropriateness, and contextual accuracy. These challenges illustrate the complexity of integrating large language models into educational environments and emphasize the importance of continuous testing and iterative design.

Overall, HexiLexi achieved its primary goals; supporting children in identifying and understanding unfamiliar vocabulary, encouraging active engagement with word meanings instead of passive reception of definitions, and promoting word consciousness and reflective thinking as foundations for lifelong literacy development. Within the framework of STREEN and LeseLab, HexiLexi represents a meaningful step toward a more interactive, adaptive, and learner-centred digital reading environment. The project demonstrates that AI can serve not merely as an information provider but as a guided learning companion that scaffolds children’s meaning-making processes.

5.2 Implications

The results of this project carry several implications for educational technology, reading instruction, and AI integration in primary education.

First, the strong positive reception of HexiLexi's visual design and gamified elements underscores the importance of aesthetics and engagement in digital learning tools for children. Motivation in young learners is closely connected to enjoyment, clarity, and perceived autonomy. The integration of points, interactive features, and voice input illustrates how multimodal interaction can enhance learner involvement.

Second, the variability in perceived comprehensibility highlights the necessity of adaptive scaffolding. Not all learners benefit equally from the same type of explanation. AI-based systems must therefore move beyond static explanations and incorporate differentiated support strategies tailored to individual learning needs, reading levels, and prior vocabulary knowledge.

Third, the project confirms that user-centred design is essential when developing educational technologies for children. Observations, questionnaires, and poster-based qualitative feedback provided valuable insights that directly informed system improvements. This iterative, participatory approach ensures that technological innovation remains grounded in real classroom experiences rather than purely theoretical assumptions.

Finally, the integration of HexiLexi into the broader STREEN framework suggests that vocabulary support should not be treated as an isolated component but as an embedded, context-sensitive feature within digital reading platforms. Vocabulary learning is most effective when directly connected to meaningful text engagement, rather than detached from authentic reading experiences.

5.3 Future Work

Although HexiLexi has shown promising initial results, further development and research are required to fully realize its potential. Future work can be structured along four main dimensions: technical refinement, pedagogical enhancement, personalization and accessibility, and empirical evaluation.

5.3.1 Technical Refinement and System Stability

One of the primary priorities for future development is ensuring long-term system stability. During testing, multiple rapid button clicks triggered excessive AI requests, occasionally leading to system crashes. While rate limiting and request control mechanisms were implemented, future versions should include more robust backend infrastructure, improved request management, and enhanced error-handling mechanisms.

Additionally, further optimization of AI prompts is necessary. Some responses were occasionally too complex or insufficiently contextualized for the target age group. Future

iterations should include structured prompt engineering strategies, age-adaptive response filtering, and systematic evaluation of AI output quality.

Another important enhancement involves the integration of visual explanations, such as images or simple diagrams, to complement textual definitions. Multimodal support may particularly benefit learners with lower reading proficiency or those who process information more effectively through visual cues.

5.3.2 Pedagogical Enhancement and Learning Design

Future work should deepen the pedagogical framework underlying HexiLexi. While the current learning modes provide varied strategies (direct explanation, guided discovery, synonyms, and word parts), further alignment with evidence-based vocabulary instruction models could strengthen learning outcomes.

Possible enhancements include:

- *contextualized example sentences tailored to the specific story being read*
- *adaptive hint sequences that gradually reduce scaffolding as competence increases*
- *reflection prompts encouraging learners to use newly acquired words in their own sentences*
- *integration of formative assessment mechanisms to track vocabulary mastery over time*

Moreover, gamification elements could be expanded through badges, levels, narrative progression, or collaborative challenges between learners. However, such features should remain pedagogically meaningful and not distract from cognitive engagement with vocabulary.

5.3.3 Personalization, Accessibility, and Inclusion

A key direction for future research lies in personalization. AI-based systems are uniquely positioned to adapt to individual learning trajectories. Future versions of HexiLexi could analyze user interaction patterns and adjust explanation complexity, hint frequency, and feedback style accordingly.

Speech interaction also offers promising potential. Improving voice recognition quality and accessibility features would make the tool more inclusive for learners with reading difficulties or special educational needs. Additional features such as adjustable text size, simplified language modes, and multilingual support could further increase accessibility.

Furthermore, integrating emotion-sensitive feedback mechanisms may enhance motivational support. Detecting patterns of frustration or disengagement (e.g., repeated incorrect responses or inactivity) could allow the system to provide encouragement or simplified assistance.

5.3.4 Long-Term Empirical Evaluation

Perhaps the most crucial step for future work is conducting systematic, long-term research. The current project was limited to short-term sessions and descriptive statistics. To draw stronger conclusions about learning effectiveness, future studies should include:

- *pre–post vocabulary assessments*
- *control group comparisons*
- *validated motivation scales*
- *longitudinal designs across several weeks or months*

Such studies would enable researchers to evaluate not only immediate usability but also sustained vocabulary acquisition, transfer to reading comprehension, and long-term motivational impact.

In addition, integrating HexiLexi more deeply into the STREEN environment would allow investigation of how vocabulary support interacts with broader reading motivation strategies. This holistic perspective aligns with the original vision of digitally augmented reading instruction and would contribute valuable insights to the field of AI-enhanced literacy education.

5.4 Final Reflection

HexiLexi demonstrates how integrating artificial intelligence into user-centred design can actively support children's literacy development. By empowering children to actively explore unfamiliar words rather than passively receiving definitions, the project promotes deeper engagement with language and supports the development of critical thinking skills.

Although further refinement and empirical validation are required, the initial findings suggest that AI can function as a supportive “word detective” companion within digital reading environments. When thoughtfully designed and pedagogically grounded, such systems have the potential to enrich vocabulary learning, strengthen reading comprehension, and contribute to more adaptive, motivating literacy instruction in primary education.

In this sense, HexiLexi not only builds upon the foundation laid by STREEN and LeseLab but also opens new pathways for research and development at the intersection of artificial intelligence and educational practice.

6. Acknowledgements

We want thank M. Sc. Pedro Ribeiro especially for guiding and supporting us throughout this whole interdisciplinary project. Also, we would like to thank Dr. Kyra Kannen, who supported the psychological research, helped us with the statistical evaluations and the poster. Furthermore, we appreciate the AIS Lab for providing the development environment, GitHub repositories and coding facilities. A special thank you goes out to the Ernst-Reuter-Grundschule, which provided us with the LeseLab and all our participants, so we were able to host our testing sessions. Finally, we want to thank Rhine – Waal – University, for making this project possible and supporting our study.

[Ribeiro et al., 2020] inspired our methodology.

7. References

7.1 Project Resources & Infrastructure

- *HexiLexi Source Code Repository: AIS Lab GitLab*, <https://git.ais-lab.eu/streen/streen-worddetective>
- *STREEN Digital Reading Platform*: <https://streen-app.ais-lab.de/>
- *AIS Lab LLM Infrastructure: API Documentation and Hosting*, <https://llm.ais-lab.eu/>

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7.3 Technologies, Frameworks, & Models:

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- Google DeepMind (2024). *Gemma: Open Models Based on Gemini Research and Technology*.
- Jamovi: <https://jamovi.org>
- Node.js: <https://nodejs.org>
- Ollama: <https://ollama.com>

- React: <https://react.dev>
- React Testing Library: <https://testing-library.com/docs/react-testing-library/intro/>
- W3C Web Speech API Specification: <https://wicg.github.io/speech-api/>

8. Contributions

The successful execution of the HexiLexi project was the result of a collaborative, interdisciplinary team effort. The specific roles and contributions of each team member are outlined below:

- **Taha Asif (Project Lead & Lead Developer):** Spearheaded the software development process, technical architecture, and system integration. Responsible for programming the React frontend, integrating the LLM via the AIS Lab infrastructure, managing the prompt engineering pipeline, and resolving technical bottlenecks. Authored Section 3 (*Implementation*) and Section 7 (*References*).
- **Emily Heye (Co-Project Lead & Lead Researcher):** Directed the pedagogical research and user-centered evaluation. Orchestrated the primary school testing sessions, designed the UX and motivation questionnaires, and conducted the subsequent data evaluation. Facilitated cross-functional communication by reporting progress between the technical and research sub-teams. Authored Section 1 (*Introduction*) and Section 6 (*Acknowledgments*).
- **Noah Schönwetter (Educational Researcher & Content Developer):** Contributed to the research and the execution of the user-centered school sessions. Designed and developed the narrative stories utilized for testing HexiLexi's contextual learning capabilities in live environments and participated in beta testing of new HexiLexi versions. Authored Section 2 (*Methodology*).
- **Leon Deda (UI Design & Data Analyst):** Assisted in the processing, structuring, and interpretation of the quantitative and qualitative data, programming the initial structure of the project, and the user interface and mascot design. Authored Section 4 (*Results*), detailing the statistical and observational findings of the project.
- **Muhammad Sufyan (Systems Analyst):** Contributed to the project's conceptual evaluation and strategic planning. Synthesized the project outcomes, identified system limitations, and outlined actionable steps for subsequent development phases. Authored Section 5 (*Conclusion and Future Work*).